2.55 Proof by Induction: General

Worked Solutions

1 a)

Prove true for $n = 1$:

RHS: $\cos 1\theta + i \sin 1\theta$ which is the LHS so true for $n = 1$

Assume true for $n = k$:

$$(\cos\theta + i\sin\theta)^k = \cos k\theta + i\sin k\theta$$

Prove true for $n = k + 1$:

$$\begin{aligned}(\cos\theta + i\sin\theta)^{k+1} &= (\cos\theta + i\sin\theta)(\cos\theta + i\sin\theta)^k \\ &= (\cos\theta + i\sin\theta)(\cos k\theta + i\sin k\theta) \\ &= \cos\theta\cos k\theta - \sin\theta\sin k\theta + i\sin\theta\cos k\theta + i\cos\theta\sin k\theta \\ &= (\cos(\theta + k\theta)) + i(\sin(\theta + k\theta)) \\ &= (\cos(k+1)\theta) + i(\sin(k+1)\theta)\end{aligned}$$

Which is $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

1 b)

$$\frac{1}{\cos\theta + i\sin\theta} \times \frac{\cos\theta - i\sin\theta}{\cos\theta - i\sin\theta} = \frac{\cos\theta - i\sin\theta}{\cos^2\theta + \sin^2\theta} = \cos\theta - i\sin\theta$$

Hence

$$\frac{1}{(\cos\theta + i\sin\theta)^n} = (\cos\theta - i\sin\theta)^n$$

But since $(\cos x + i\sin x)^n = \cos nx + i\sin nx$ then set $x = -\theta$ so that

$$(\cos(-\theta) + i\sin(-\theta))^n = \cos n(-\theta) + i\sin n(-\theta)$$

$$\Rightarrow (\cos\theta - i\sin\theta)^n = \cos n\theta - i\sin n\theta$$

So that $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ for all integers $n < 0$

When $n = 0$:

$$(\cos\theta + i\sin\theta)^0 = 1 \text{ and } \cos 0 + i\sin 0 = 1$$

$$\therefore (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta \text{ for all integers } n$$

2)

Prove true for $n = -1$:

Note that:

$$\frac{1}{\cos\theta + i\sin\theta} \times \frac{\cos\theta - i\sin\theta}{\cos\theta - i\sin\theta} = \frac{\cos\theta - i\sin\theta}{\cos^2\theta + \sin^2\theta} = \cos\theta - i\sin\theta$$

RHS: $\cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta = \frac{1}{\cos\theta + i\sin\theta}$ which is the LHS so true for $n = -1$

Assume true for $n = k$:

$$(\cos\theta + i\sin\theta)^k = \cos k\theta + i\sin k\theta$$

Prove true for $n = k - 1$:

$$\begin{aligned}(\cos\theta + i\sin\theta)^{k-1} &= (\cos\theta + i\sin\theta)^k(\cos\theta - i\sin\theta)\\ &= (\cos k\theta + i\sin k\theta)(\cos\theta - i\sin\theta)\\ &= \cos k\theta\cos\theta + \sin k\theta\sin\theta + i(\sin k\theta\cos\theta - \cos k\theta\sin\theta)\\ &= (\cos(k\theta - \theta)) + i(\sin(k\theta - \theta))\\ &= (\cos(k-1)\theta) + i(\sin(k-1)\theta)\end{aligned}$$

Which is $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ with $k - 1$ substituted in.

As the statement is true for $n = -1$, and if the statement is true for $n = k$ then the statement is true for $n = k - 1$, so the statement is true for all integers $n \leq -1$

3)

Prove true for $n = 1$:

LHS: $\frac{1}{3}$

RHS: $\frac{1}{2\times1+1} = \frac{1}{3}$ so true for $n = 1$

Assume true for $n = k$:

$$\frac{1}{3} \times \frac{3}{5} \times \frac{5}{7} \times \ldots \times \frac{2k-1}{2k+1} = \frac{1}{2k+1}$$

Prove true for $n = k + 1$:

$$\begin{aligned}\frac{1}{3} \times \frac{3}{5} \times \frac{5}{7} \times \ldots \times \frac{2k-1}{2k+1} \times \frac{2k+1}{2k+3} &= \frac{1}{2k+1} \times \frac{2k+1}{2k+3}\\ &= \frac{1}{2k+3} = \frac{1}{2(k+1)+1}\end{aligned}$$

Which is $\frac{1}{2n+1}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

4)

Prove true for $n = 1$:

LHS:

$$\sum_{r=0}^{1} \binom{1}{r} = \binom{1}{0} + \binom{1}{1} = 2$$

RHS: $2^1 = 2$ so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=0}^{k} \binom{k}{r} = 2^k$$

Prove true for $n = k + 1$:

$$\sum_{r=0}^{k+1} \binom{k+1}{r}$$

Since $\binom{k+1}{r} = \binom{k}{r} + \binom{k}{r-1}$

$$\sum_{r=0}^{k+1} \binom{k+1}{r} = \sum_{r=0}^{k+1} \binom{k}{r} + \sum_{r=0}^{k+1} \binom{k}{r-1}$$

Since $\binom{k}{r-1}$ is only defined for $r \geq 1$

$$\sum_{r=0}^{k+1} \binom{k}{r-1} = \sum_{r=1}^{k+1} \binom{k}{r-1} = \sum_{r=0}^{k} \binom{k}{r}$$

So

$$\begin{aligned}\sum_{r=0}^{k+1} \binom{k+1}{r} &= \sum_{r=0}^{k+1} \binom{k}{r} + \sum_{r=0}^{k} \binom{k}{r} \\ &= \sum_{r=0}^{k} \binom{k}{r} + \sum_{r=0}^{k} \binom{k}{r} \\ &= 2^k + 2^k = 2 \times 2^k = 2^{k+1}\end{aligned}$$

So true for $n = k + 1$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

5)

Prove true for $n = 1$:

LHS:

$$\sum_{r=1}^{1} \ln(2r-1) = \ln(2-1) = \ln 1 = 0$$

$\ln\left(\frac{(2n)!}{2^n n!}\right) = \ln\left(\frac{2!}{2^1 \times 1!}\right) = \ln\left(\frac{2}{2}\right) = \ln 1 = 0$ so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \ln(2r-1) = \ln\left(\frac{(2k)!}{2^n k!}\right)$$

Prove true for $n = k + 1$:

$$\begin{aligned}\sum_{r=1}^{k+1} \ln(2r-1) &= \sum_{r=1}^{k} \ln(2r-1) + \ln(2(k+1)-1) \\ &= \ln\left(\frac{(2k)!}{2^k k!}\right) + \ln(2k+1) \\ &= \ln\left(\frac{(2k+1)(2k)!}{2^k k!}\right) \\ &= \ln\left(\frac{(2k+2)(2k+1)(2k)!}{(2k+2)2^k k!}\right) \\ &= \ln\left(\frac{(2k+2)!}{2(k+1)2^k k!}\right) \\ &= \ln\left(\frac{(2k+2)!}{2^{k+1}(k+1)!}\right) = \ln\left(\frac{(2(k+1))!}{2^{k+1}(k+1)!}\right)\end{aligned}$$

Which is $\ln\left(\frac{(2n)!}{2^n n!}\right)$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

6)

Prove true for $n = 1$:

LHS: $\mathrm{f}^1(x) = \frac{x}{x+1}$

RHS: $\frac{x}{1x+1} = \frac{x}{x+1}$ so true for $n = 1$

Assume true for $n = k$:

$$\mathrm{f}^k(x) = \frac{x}{kx+1}$$

Prove true for $n = k + 1$:

$$\begin{aligned}\mathrm{f}^{k+1}(x) = \mathrm{f}(\mathrm{f}^k(x)) &= \mathrm{f}\left(\frac{x}{kx+1}\right) \\ &= \frac{\frac{x}{kx+1}}{\frac{x}{kx+1}+1} = \frac{x}{x+kx+1} = \frac{x}{(k+1)x+1}\end{aligned}$$

which $\frac{x}{kx+1}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

7)

Prove true for $n = 1$:

LHS: $\frac{d}{dx}(x^1) = 1$

RHS: $n! = 1! = 1$ so true for $n = 1$

Assume true for $n = k$:

$$\frac{d^k}{dx^k}(x^k) = k!$$

Prove true for $n = k + 1$:

$$\begin{aligned}\frac{d^{k+1}}{dx^{k+1}}(x^{k+1}) &= \frac{d}{dx}\left(\frac{d^k}{dx^k}(x^{k+1})\right) \\ &= \frac{d}{dx}((k+1)x^k) = (k+1)\frac{d}{dx}(x^k) \\ &= (k+1)k! = (k+1)! \quad \text{so true for } n = k+1\end{aligned}$$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

8 a)

Prove true for $n = 1$:

LHS: $\frac{d}{d\theta}(\sin a\theta) = a\cos a\theta$

RHS: $a\sin\left(a\theta + \frac{\pi}{2}\right)$

Since $\sin\left(x + \frac{\pi}{2}\right) = \cos x$

Let $x = a\theta$ so $a\sin\left(a\theta + \frac{\pi}{2}\right) = a\cos a\theta$ so true for $n = 1$

Assume true for $n = k$:

$$\frac{d^k}{d\theta^k}(\sin a\theta) = a^k \sin\left(a\theta + \frac{k\pi}{2}\right)$$

Prove true for $n = k + 1$:

$$\frac{d^{k+1}}{d\theta^{k+1}}(\sin a\theta) = \frac{d}{d\theta}\left(\frac{d^k}{d\theta^k}(\sin a\theta)\right)$$

$$
\begin{aligned}
&= \frac{d}{d\theta}\left(a^k \sin\left(a\theta + \frac{k\pi}{2}\right)\right) \\
&= a^{k+1} \cos\left(a\theta + \frac{k\pi}{2}\right) \\
&= a^{k+1} \sin\left(a\theta + \frac{k\pi}{2} + \frac{\pi}{2}\right) \\
&= a^{k+1} \sin\left(a\theta + \frac{(k+1)\pi}{2}\right)
\end{aligned}
$$

Which is $a^n \sin\left(a\theta + \frac{n\pi}{2}\right)$ with $n = k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$

8 b)

Prove true for $n = 1$:

LHS: $\frac{d}{d\theta}(\ln(1+ax)) = \frac{a}{1+ax}$

RHS: $\frac{(-1)^{1+1}(1-1)!a^1}{(1+ax)^1} = \frac{a}{1+ax}$ so true for $n = 1$

Assume true for $n = k$:

$$\frac{d^k}{dx^k}(\ln(1+ax)) = \frac{(-1)^{k+1}(k-1)!a^k}{(1+ax)^k}$$

Prove true for $n = k+1$:

$$
\begin{aligned}
\frac{d^{k+1}}{dx^{k+1}}(\ln(1+ax)) &= \frac{d}{dx}\left(\frac{d^k}{dx^k}(\ln(1+ax))\right) \\
&= \frac{d}{dx}\left(\frac{(-1)^{k+1}(k-1)!a^k}{(1+ax)^k}\right) \\
&= \frac{(-1)^{k+1}(k-1)!a^k \times -ka}{(1+ax)^k} \\
&= \frac{(-1)(-1)^{k+1} \times k(k-1)! \times a \times a^k}{(1+ax)^k} \\
&= \frac{(-1)^{k+2}k!a^{k+1}}{(1+ax)^{k+1}} \\
&= \frac{(-1)^{(k+1)+1}(k+1-1)!a^{k+1}}{(1+ax)^{k+1}}
\end{aligned}
$$

Which is $\frac{(-1)^{n+1}(n-1)!a^n}{(1+ax)^n}$ with $n = k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$

9 a)

Prove true for $n = 1$:

LHS: $\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$

RHS: $a^1 e^{ax+b}$ so true for $n = 1$

Assume true for $n = k$:

$$\frac{d^k}{dx^k}(e^{ax+b}) = a^k e^{ax+b}$$

Prove true for $n = k + 1$:

$$\begin{aligned}\frac{d^{k+1}}{dx^{k+1}}(e^{ax+b}) &= \frac{d}{dx}\left(\frac{d^k}{dx^k}(e^{ax+b})\right) \\ &= \frac{d}{dx}(a^k e^{ax+b}) = a^{k+1}e^{ax+b}\end{aligned}$$

Which is $a^n e^{ax+b}$ with $n = k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

9 b)

Let $\mathrm{f}(x) = e^{ax+b}$, so $\mathrm{f}(0) = e^b$

Let $\mathrm{f}'(x) = ae^{ax+b}$, so $\mathrm{f}'(0) = ae^b$

Let $\mathrm{f}''(x) = a^2e^{ax+b}$, so $\mathrm{f}''(0) = a^2e^b$

$$\therefore e^{ax+b} = e^b + ae^b x + \frac{a^2e^b}{2}x^2$$

10)

Prove true for $n = 1$:

LHS:

$$\begin{aligned}\frac{d}{dx}(e^x\cos(\sqrt{3}x)) &= e^x\cos(\sqrt{3}x) - \sqrt{3}e^x\sin(\sqrt{3}x) \\ &= e^x[\cos(\sqrt{3}x) - \sqrt{3}\sin(\sqrt{3}x)] \\ &= 2e^x\left[\frac{1}{2}\cos(\sqrt{3}x) - \frac{\sqrt{3}}{2}\sin(\sqrt{3}x)\right] \\ &= 2e^x\left[\cos\left(\frac{\pi}{3}\right)\cos(\sqrt{3}x) - \sin\left(\frac{\pi}{3}\right)\sin(\sqrt{3}x)\right] \\ &= 2e^x\cos\left(\frac{\pi}{3} + \sqrt{3}x\right) = 2e^x\cos\left(\sqrt{3}x + \frac{\pi}{3}\right)\end{aligned}$$

RHS: $2^1e^x\cos\left(\sqrt{3}x + \frac{\pi}{3} \times 1\right)$ so true for $n = 1$

Assume true for $n = k$:

$$\frac{d^k}{dx^k}(e^x\cos(\sqrt{3}x)) = 2^k e^x\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right)$$

Prove true for $n = k + 1$:

$$\begin{aligned}\frac{d^{k+1}}{dx^{k+1}}(e^x\cos(\sqrt{3}x)) &= \frac{d}{dx}\left(\frac{d^k}{dx^k}(e^x\cos(\sqrt{3}x))\right) \\ &= \frac{d}{dx}\left(2^k e^x\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right)\right) \\ &= -2^k e^x \times \sqrt{3}\sin\left(\sqrt{3}x + \frac{\pi}{3}k\right) + 2^k e^x\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right) \\ &= 2^k e^x\left[\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right) - \sqrt{3}\sin\left(\sqrt{3}x + \frac{\pi}{3}k\right)\right] \\ &= 2 \times 2^k e^x\left[\frac{1}{2}\cos\left(\sqrt{3}x + \frac{\pi}{3}k\right) - \frac{\sqrt{3}}{2}\sin\left(\sqrt{3}x + \frac{\pi}{3}k\right)\right]\end{aligned}$$

$$= 2^{k+1}e^x\left[\cos\left(\frac{\pi}{3}\right)\cos\left(\sqrt{3}x+\frac{\pi}{3}k\right)-\sin\left(\frac{\pi}{3}\right)\sin\left(\sqrt{3}x+\frac{\pi}{3}k\right)\right]$$
$$= 2^{k+1}e^x\left[\cos\left(\frac{\pi}{3}+\left(\sqrt{3}x+\frac{\pi}{3}k\right)\right)\right]$$
$$= 2^{k+1}e^x\left[\cos\left(\sqrt{3}x+\frac{\pi}{3}(k+1)\right)\right]$$

Which is $2^n e^x \cos\left(\sqrt{3}x+\frac{\pi}{3}n\right)$ with $n=k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n \geq 1$

11)

Prove true for $n=2$:

LHS:

$$\frac{d}{dx}\left(\frac{d}{dx}(\sinh x)\right) = \frac{d}{dx}(\cosh x) = \sinh x$$

which is the RHS so true for $n=2$

Assume true for some even integer $n=k$ where $k \geq 2$:

$$\frac{d^k y}{dx^k} = \frac{d^k}{dx^k}(\sinh x) = \sinh x$$

Prove true for $n=k+2$:

$$\frac{d^{k+2}}{dx^{k+2}}(\sinh x) = \frac{d^2}{dx^2}\left(\frac{d^k}{dx^k}(\sinh x)\right)$$
$$= \frac{d^2}{dx^2}(\sinh x) = \frac{d}{dx}\left(\frac{d}{dx}(\sinh x)\right)$$
$$= \frac{d}{dx}(\cosh x) = \sinh x$$

As the statement is true for $n=2$, and if the statement is true for $n=k$ then the statement is true for $n=k+2$, so the statement is true for all even integers $n \geq 2$

12)

Prove true for $n=1$:

LHS: $\frac{d}{dx}(x\sinh x) = x\cosh x + \sinh x$

RHS: $x\cosh x + (2-1)\sinh x = x\cosh x + \sinh x$ so true for $n=1$

Assume true for $n=k$:

$$\frac{d^{2k-1}y}{dx^{2k-1}} = x\cosh x + (2k-1)\sinh x$$

Prove true for $n=k+1$:

$$\frac{d^{2(k+1)-1}y}{dx^{2(k+1)-1}} = \frac{d^{2k+1}y}{dx^{2k+1}}$$
$$= \frac{d^2}{dx^2}\left(\frac{d^{2k-1}y}{dx^{2k-1}}\right)$$
$$= \frac{d^2}{dx^2}(x\cosh x + (2k-1)\sinh x)$$
$$= \frac{d}{dx}(x\sinh x + \cosh x + (2k-1)\cosh x)$$

$$= \frac{d}{dx}(x\sinh x + 2k\cosh x)$$
$$= x\cosh x + \sinh x + 2k\sinh x = x\cosh x + (2k+1)\sinh x$$
$$= x\cosh x + (2(k+1)-1)\sinh x$$

Which is $x\cosh x + (2n-1)\sinh x$ with $n = k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$

13)

Prove true for $n = 1$:

LHS: $\frac{d}{dx}(e^x\cosh x) = e^x\sinh x + e^x\cosh x = e^x(\cosh x + \sinh x)$

RHS: $2^0e^x(\cosh x + \sinh x) = e^x(\cosh x + \sinh x)$ so true for $n = 1$

Assume true for $n = k$:

$$\frac{d^ky}{dx^k} = 2^{k-1}e^x(\cosh x + \sinh x)$$

Prove true for $n = k+1$:

$$\frac{d^{k+1}y}{dx^{k+1}} = \frac{d}{dx}\left(\frac{d^ky}{dx^k}\right)$$
$$= \frac{d}{dx}(2^{k-1}e^x(\cosh x + \sinh x))$$
$$= 2^{k-1}e^x(\sinh x + \cosh x) + 2^{k-1}e^x(\cosh x + \sinh x)$$
$$= 2\times 2^{k-1}e^x(\cosh x + \sinh x) = 2^ke^x(\cosh x + \sinh x)$$
$$= 2^{(k+1)-1}e^x(\cosh x + \sinh x)$$

Which is $2^{n-1}e^x(\cosh x + \sinh x)$ with $n = k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$

14)

Prove true for $n = 1$:

LHS: x

RHS: $\frac{x(1-(-x)^1)}{1+x} = \frac{x(1+x)}{1+x} = x$ so true for $n = 1$

Assume true for $n = k$:

$$f_k(x) = \frac{x(1-(-x)^k)}{1+x}$$

Prove true for $n = k+1$:

$$f_{k+1}(x) = f_k(x) + (-1)^{k+1-1}x^{k+1}$$
$$= \frac{x(1-(-x)^k)}{1+x} + (-1)^kx^{k+1}$$
$$= \frac{x(1-(-x)^k)}{1+x} + (-x)^kx$$
$$= x\left(\frac{1-(-x)^k}{1+x} + (-x)^k\right)$$
$$= x\left(\frac{1-(-x)^k}{1+x} + \frac{(1+x)(-x)^k}{1+x}\right)$$

$$
\begin{aligned}
&= x\left(\frac{1-(-x)^k+(1+x)(-x)^k}{1+x}\right)\\
&= x\left(\frac{1-(-x)^k+(-x)^k+x(-x)^k}{1+x}\right)\\
&= x\left(\frac{1+x(-x)^k}{1+x}\right) = x\left(\frac{1-(-x)^{k+1}}{1+x}\right)\\
&= \frac{x(1-(-x)^{k+1})}{1+x}
\end{aligned}
$$

Which is $\frac{x(1-(-x)^n)}{1+x}$ with $n = k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$

15)

Prove true for $n = 1$:

LHS: $(1+x)^1 = 1+x$

RHS: $1 + {}^1C_1 x = 1+x$ so true for $n = 1$

Assume true for $n = k$:

$$(1+x)^k \equiv 1 + {}^kC_1 x + {}^kC_2 x^2 + \cdots + {}^kC_k x^k$$

Prove true for $n = k+1$:

$$
\begin{aligned}
(1+x)^{k+1} &= (1+x)(1+x)^k\\
&= (1+x)(1+{}^kC_1 x + {}^kC_2 x^2 + \cdots + {}^kC_k x^k)\\
&= 1 + {}^kC_1 x + {}^kC_2 x^2 + \cdots + {}^kC_k x^k\\
&\quad + x + {}^kC_1 x^2 + {}^kC_2 x^3 + \cdots + {}^kC_k x^{k+1}\\
&= 1 + (1+{}^kC_1)x + ({}^kC_1 + {}^kC_2)x^2 + ({}^kC_2 + {}^kC_3)x^3\\
&\quad + \cdots + ({}^kC_{k-1} + {}^kC_k)x^k + ({}^kC_k)x^{k+1}
\end{aligned}
$$

But Pascal's Rule gives: ${}^kC_{r-1} + {}^kC_r = {}^{k+1}C_r$

So that

$$(1+x)^{k+1} = 1 + {}^{k+1}C_1 x + {}^{k+1}C_2 x^2 + {}^{k+1}C_3 x^3 + \cdots + {}^{k+1}C_k x^k + ({}^{k+1}C_{k+1})x^{k+1}$$

Which is $(1+x)^n \equiv 1 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$ with $n = k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$